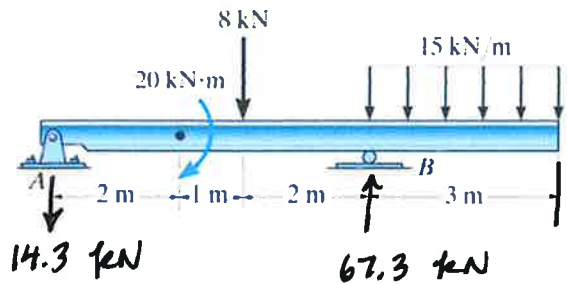
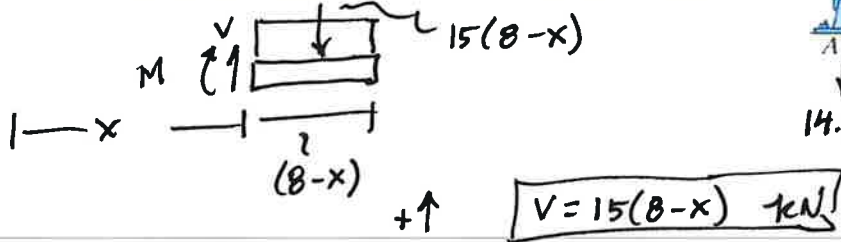


1.5 Write the shear equation for the segment between  $5 < x < 8$  (origin of x-axis is at point A).

Support Reactions  $A_y = 14.3 \text{ kN} \downarrow$  and  $B_y = 67.3 \text{ kN} \uparrow$



1.6 Write the moment equation for the segment between  $5 < x < 8$  (origin of x-axis is at point A)

Support Reactions  $A_y = 14.3 \text{ kN} \downarrow$   $B_y = 67.3 \text{ kN} \uparrow$

$$+\circlearrowleft \sum M_{\text{cut}} = 0; \quad M = -15(8-x)\left(\frac{8-x}{2}\right)$$

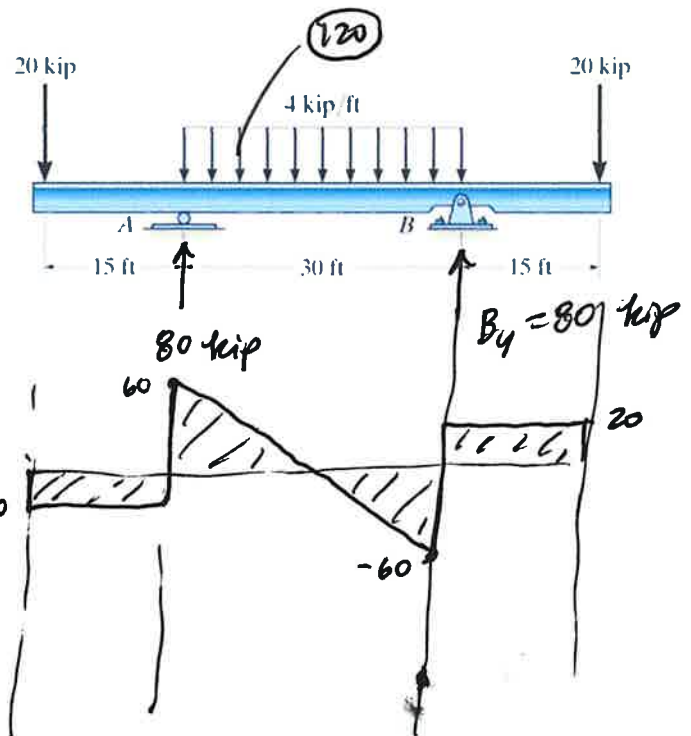
$$M = -7.5(8-x)^2$$

## Section 2 – Graphing VM Diagrams with 6 rules

Support reactions.....

$A_y = B_y = 80 \text{ kip} \uparrow$

The x axis has its origin at the left side of the beam



2.1 What is the numerical value of change (magnitude) of the change in shear for the section  $15 < x < 45 \text{ ft}$  and what is the line type for the connection?

-120 kip



2.2 What happens to the shear diagram at  $x = 45 \text{ ft}$ ?

Jumps  $\uparrow$   $80 \text{ kip}$